

CASE STUDY · MANUFACTURING · DEMAND PLANNING

Forecasting Boat Demand, Season After Season

A SKU-level Python forecasting system for White River Marine that lifted accuracy, cut stockouts, and freed up working capital tied to inventory.

12–18%

MAPE IMPROVEMENT

10–15%

STOCKOUT REDUCTION

8–12%

INVENTORY TURNS LIFT

3+ yrs

HISTORY MODELED

Boats are highly seasonal — and we kept missing the season.

THE PRODUCT

White River Marine manufactures recreational fishing boats sold through a network of dealers. Demand peaks hard in spring and summer — and missed peak demand is sales we can't recover.

HOW WE WERE PLANNING

Forecasts were built in spreadsheets — judgmental, regional estimates with no consistent method. Error rates were high and varied wildly across SKUs.



Stockouts on hot SKUs

Top-selling models ran out at key dealers early in the season — every empty lot was a customer who walked.



Capital trapped in slow movers

Other models sat in dealer yards for months, tying up working capital and dragging on inventory turns.



Inconsistent SKU-level signal

Spreadsheet forecasts varied by analyst, region, and season — there was no defensible system to compare against.

Move from spreadsheet judgment to SKU-level statistical forecasts.

THE PROPOSAL

Build a Python time-series forecasting system that produces monthly SKU × region forecasts with confidence intervals — and feeds them back into the Sales & Ops planning cycle.



GRANULARITY

Monthly forecasts at SKU × region — small enough to be useful for allocation, large enough to be statistically stable.

Co-defined with Sales & Ops



HORIZON

Aligned with manufacturing & shipping lead times — far enough out for Operations to actually act on the signal.

Decision-window aware



PRIMARY METRICS

Forecast MAPE, stockout rate, and inventory turns — accuracy plus the operational outcomes we actually cared about.

Business outcomes, not RMSE



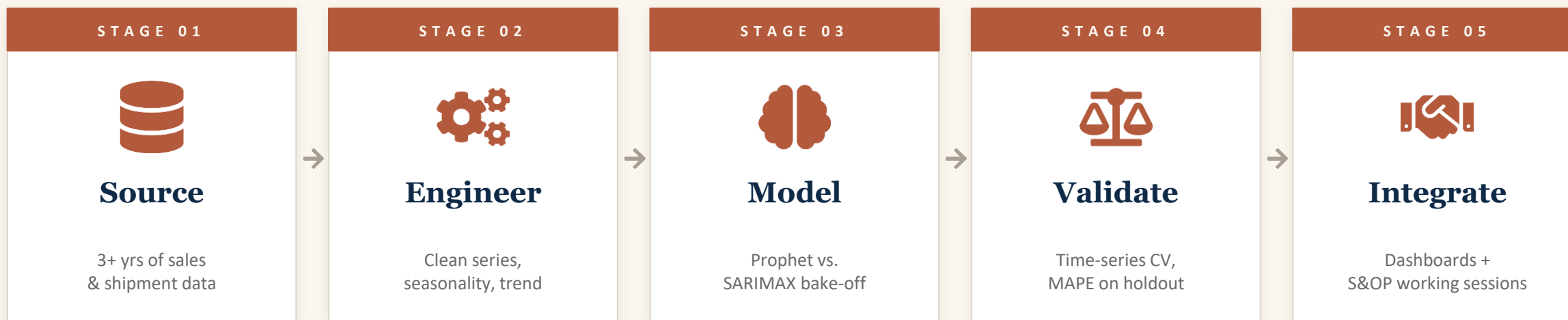
STAKEHOLDERS

Sales for regional knowledge; Operations for build/ship feasibility; Finance for capital impact framing.

Cross-functional from day one

Five stages — from raw shipment data to recurring planning rhythm.

I owned the lifecycle: data, modeling, evaluation, and the integration that turned model output into actual planning conversations with Sales & Ops.



WHY ALL FIVE STAGES MATTERED

Skipping integration means a great forecast that nobody uses. Sales & Ops working sessions were as critical to impact as the model itself.

Three years of history, hundreds of SKUs, one consistent time series.

DATA PULLED



Sales & shipments

3+ years of historical sales and shipment data across hundreds of SKUs and dealers.



SKU × region grid

Built a consistent monthly time series per SKU/region — the unit of forecast.



Lifecycle events

New product launches, end-of-life models, and discontinuations — explicitly tracked, not silently buried.

CLEANING DECISIONS

1

Consistent time index

Filled gaps so every SKU × region had a regular monthly cadence — no implicit zeros, no missing months.

2

Outlier handling

Spikes from one-off events (recalls, promo bursts) flagged or down-weighted to avoid distorting trend/seasonality fits.

3

Lifecycle adjustments

Newly-launched SKUs got short-history modeling; EOL SKUs were excluded after their cutoff to avoid biasing the future.

PRINCIPLE · Forecast quality is set in cleaning, not modeling. Garbage time-series in = garbage forecasts out.

Three patterns shaped every modeling decision that followed.

Before any modeling, I sat with 3 years of monthly data and let the patterns drive the design.



Strong annual seasonality

WHAT WE SAW

- Spring/summer peaks 3–5× winter troughs
- Pattern repeats year-over-year
- Holiday and weather effects layered on top

IMPLICATION: Demanded a model that could fit annual cycles cleanly — Prophet's Fourier seasonality and SARIMAX's seasonal terms were both viable.



Trend by product line

WHAT WE SAW

- Some lines growing year-over-year
- Others mature or in decline
- Trend shifts at product refresh points

IMPLICATION: Models needed flexible trend handling — Prophet's piecewise growth and changepoints, or SARIMAX's differencing, depending on SKU.



Heterogeneous SKU behavior

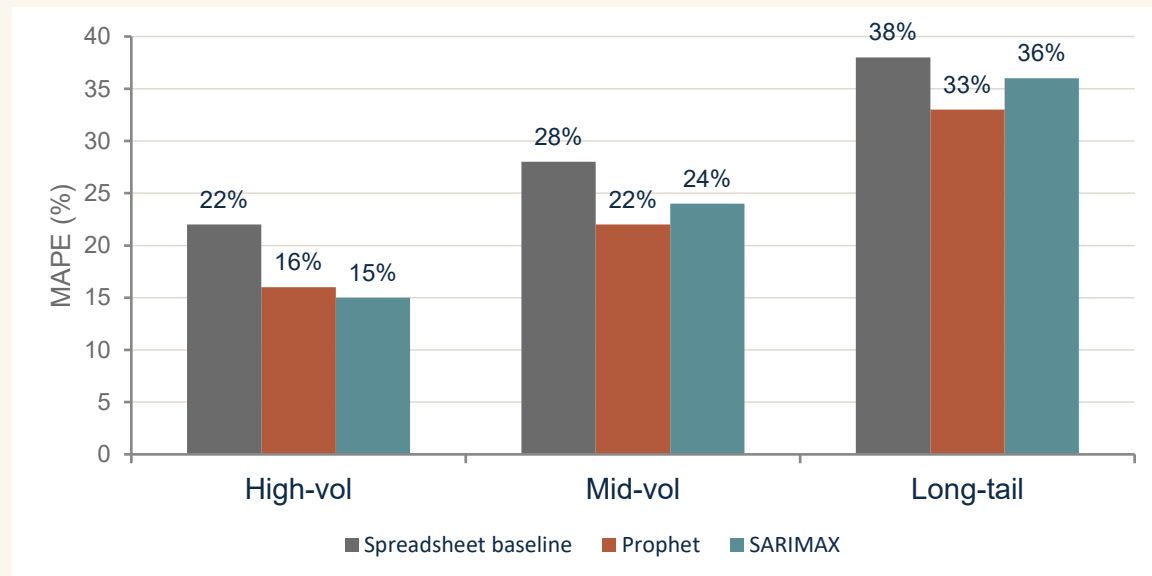
WHAT WE SAW

- High-volume SKUs: stable, model well
- Long-tail SKUs: noisy, intermittent
- Promo SKUs: distorted history

IMPLICATION: One model wouldn't fit all — pushed us toward a per-SKU model selection framework rather than a global model.

Prophet and SARIMAX competed per SKU — best-of pipeline won.

BAKE-OFF — MAPE BY SKU CLUSTER



No single model won across the board — Prophet edged ahead on mid-volume SKUs, SARIMAX on the noisier high-vol ones.

PER-SKU MODEL SELECTION



Prophet

Piecewise-linear or logistic growth + Fourier seasonality + holiday effects. Strong on lines with clear trend changes and rich seasonality.



SARIMAX

Classical ARIMA(p,d,q)×(P,D,Q)_s with optional exogenous regressors. Strong on stationary, well-behaved high-volume SKUs.



Best-of selection

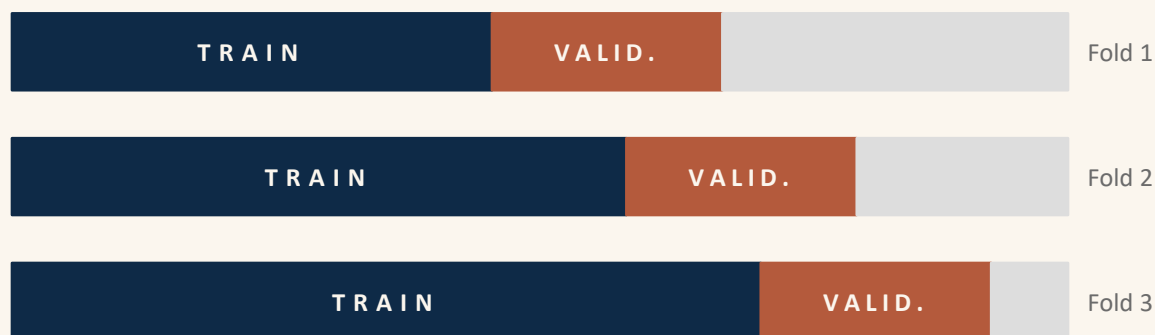
Per SKU (or SKU cluster), CV picked the better performer. Output: a forecast + interval per SKU × region from whichever model won.

PRINCIPLE • One model rarely wins across a heterogeneous portfolio. A per-SKU bake-off framework beat picking a single "global" model.

Time-series cross-validation — no peeking, ever.

ROLLING-ORIGIN VALIDATION

Pretend we're standing at successively later points in time — train only on what we'd have known, then forecast forward. Repeat. Average.



METRICS WE OPTIMIZED



MAPE — primary

Mean Absolute Percentage Error. Interpretable as "how far off, on average, in %."



Bias check

Watched mean error to catch systematic over/under-forecasting that MAPE alone hides.



Stockout rate (downstream)

Translated forecast error into operational outcome — what Sales & Ops actually felt.



Inventory turns (downstream)

Capital efficiency lens — measured on the receiving end of the planning cycle.

GUARDRAIL • No look-ahead bias. Each fold's training data is strictly in the past relative to its validation window.

From forecast tables to seasonal buy decisions.



Forecast

Per SKU × region
with intervals



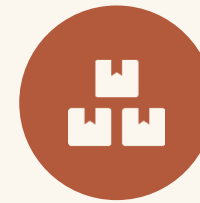
Dashboards

Expected demand
+ confidence bands



S&OP sessions

Sales adds local
knowledge



Allocation

Recommended
inventory by region



Build & ship

Operations executes
on firm plan



Dashboards Sales would actually open

Per-SKU forecast + confidence interval, regional cuts, comparison to prior year.
Built so a sales lead could glance once and know what to act on.



Human-in-the-loop adjustments

Sales had local knowledge the model didn't — competitor pricing, regional events. The process let them adjust on top of the forecast, not against it.

INTEGRATION PRINCIPLE • *The forecast was the starting point of the conversation, not the end.*

Earning Sales & Ops trust took three deliberate moves.

A statistical forecast that disagrees with regional sales judgment can feel threatening. The plan was to make it useful before making it authoritative.

1 Show the work, not just the number

Surfaced trend, seasonality, and confidence bands alongside each forecast. Sales could see why the model thought what it thought.

2 Walk through it together

Working sessions with Sales and Ops to review forecasts before each season. Their adjustments improved outputs — and built buy-in.

3 Compare every cycle

After each season, compared model forecasts vs. spreadsheet baseline vs. actuals. The track record made the case more than any deck could.

Three metrics moved together — accuracy, service level, capital efficiency.

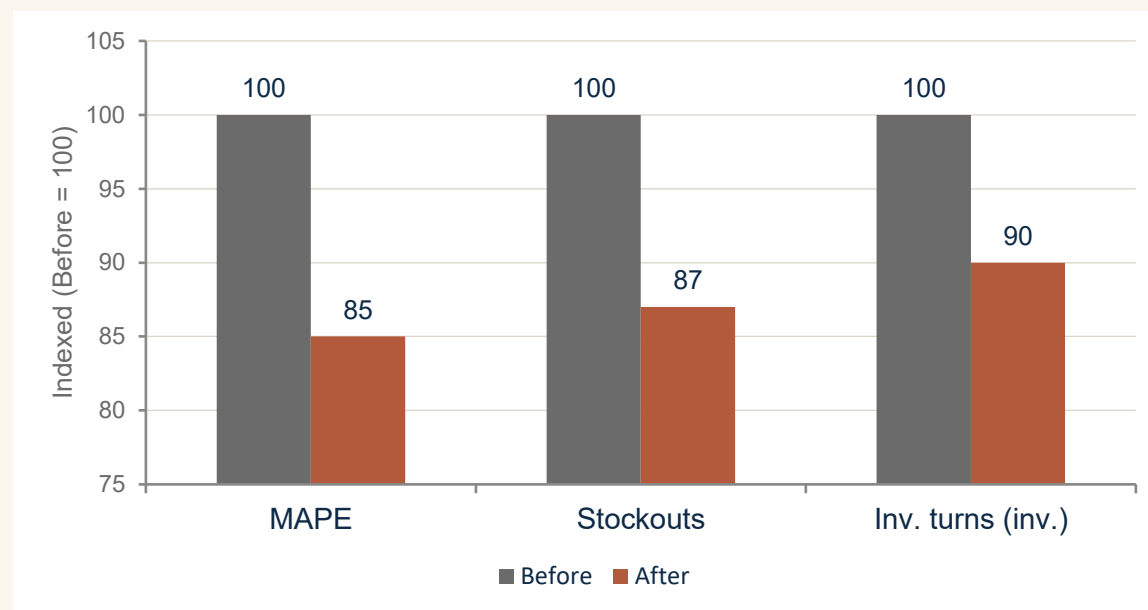
HOW WE MEASURED

Compared the new system to prior planning cycles using held-out actuals — and tracked the operational outcomes Sales & Ops actually cared about.

WHAT MOVED

- MAPE — accuracy improvement vs. baseline
- Stockout rate — service-level outcome
- Inventory turns — capital efficiency
- Sales leadership confidence in numbers

BEFORE vs. AFTER (RELATIVE)



Lower is better for all three. Inventory turns shown as inverse so direction is consistent.



HEADLINE IMPACT

-12 to -18% MAPE • **-10 to -15% stockouts** • **+8 to +12% inventory turns** across key boat lines.

The scoreboard.



MAPE IMPROVEMENT

-12 to -18%

vs. spreadsheet baseline, on held-out periods



STOCKOUT REDUCTION

-10 to -15%

fewer dealer-level stockouts in key boat lines



INVENTORY TURNS LIFT

+8 to +12%

freed working capital from idle dealer inventory



PLANNING RHYTHM

Recurring

framework now the starting point for each season's plan

QUALITATIVE WINS

Sales leadership gained confidence in the numbers · Forecasting framework became the starting point for each season's planning · Pipeline reusable for future product lines and brands.

What worked, what I'd extend next.

WHAT WORKED



Per-SKU bake-off, not one global model

Heterogeneous SKUs needed heterogeneous models. The selection framework let each line use what fit it best.



Confidence intervals as a feature, not a footnote

Bands gave Sales & Ops a way to talk about risk and where to hold safety stock — turning forecasts into planning conversations.



Working sessions before authority

Earning input rights before decision rights. The forecast didn't override Sales — it augmented them.

WHAT I'D EXTEND NEXT



Hierarchical reconciliation

Forecast at SKU × region but reconcile up to brand and total. Bottom-up + top-down constraints would tighten consistency.



Exogenous regressors at scale

Pull in macro signals — fishing licenses issued, fuel prices, weather — as covariates in SARIMAX or via Prophet's regressors.



Ensemble across models

Blend Prophet + SARIMAX + naive seasonal as a weighted ensemble per SKU for further accuracy gains.

From spreadsheet judgment to a recurring planning system.

01

Match the model to the SKU.

Heterogeneous portfolios beat heterogeneity with a selection framework, not a one-size-fits-all model.

02

Forecast accuracy is a means, not the goal.

What mattered was stockouts down, turns up, and Sales trusting the numbers — MAPE was just how we got there.

03

Build for the working session.

The forecast had to be legible enough for Sales & Ops to argue with constructively. Confidence intervals and clear cuts made it possible.

Thank you · Questions?